DSA1604 - Knowledge Engineering and Data Science

Lab Experiments

Experiment 1: Knowledge Representation Using Python

Aim:

To represent facts about a student using a Python dictionary and display the stored knowledge.

Algorithm:

1. Start the program.
2. Create a dictionary containing student details.
3. Store name, department, course, and year as key-value pairs.
4. Display the knowledge representation heading.
5. Traverse the dictionary using the items() method.
6. Print each attribute and its corresponding value.
7. Stop the program.

Pseudocode:

```
BEGIN
   CREATE student dictionary
   DISPLAY heading
   FOR each key and value in student
      PRINT key and value
   END FOR
END
```

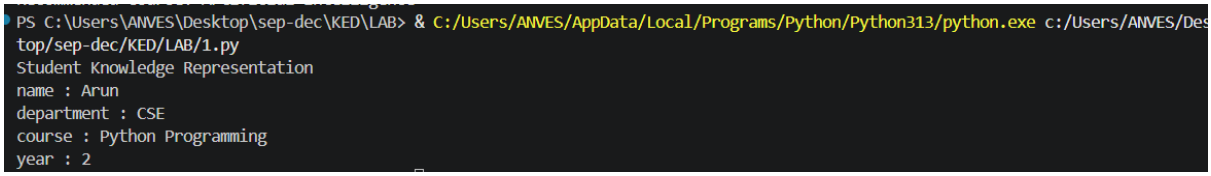
Program:

```
student = {
    "name": "Arun",
    "department": "CSE",
    "course": "Python Programming",
    "year": 2
}
print("Student Knowledge Representation")
for key, value in student.items():
    print(key, ":", value)
```

Input:

Predefined facts: name=Arun, department=CSE, course=Python Programming, year=2

Output:



Result:

Thus, student knowledge was represented using a Python dictionary and displayed successfully.

Experiment 2: Knowledge Acquisition

Aim:

To acquire knowledge from a user and store it as structured facts.

Algorithm:

1. Start the program.
2. Read the expert's name from the user.
3. Read the area of expertise.
4. Read the years of experience as an integer.
5. Store the collected information in a dictionary.
6. Display the acquired knowledge with the expert's details.
7. Stop the program.

Pseudocode:

```
BEGIN
   INPUT expert name
   INPUT area of expertise
   INPUT years of experience
   STORE details in knowledge dictionary
   DISPLAY expert name
   DISPLAY area of expertise
   DISPLAY years of experience
END
```

Program:

```
name=input("enter expert name: ")
domain=input("enter area of expertise: ")
experience=int(input("enter years of experience: "))
knowledge={
   "expert":name,
   "domain":domain,
```

```
    "experience":experience
}
print("\nAcquired Knowledge")
print("Expert:", knowledge["expert"])
print("domain:", knowledge["domain"])
print("Expertience:", knowledge["experience"], "years")
```

Input:

```
PS C:\Users\ANVES\Desktop\sep-dec\KED\LAB> & C:/Users/ANVES/AppData/Local/Programs/Python/Python313/python.exe c:/Users/ANVES/Desk
top/sep-dec/KED/LAB/2.py
enter expert name: ravi
enter area of expertise: networking
enter years of experience: 8
```

Output:

```
Acquired Knowledge
Expert: ravi
domain: networking
Expertience: 8 years
PS C:\Users\ANVES\Desktop\sep-dec\KED\LAB>
```

Result:

Thus, knowledge was acquired from the user and stored and displayed successfully in a structured form.

Experiment 3: Knowledge-Based Course Recommendation System

Aim:

To recommend a course using a small knowledge base and decision rules.

Algorithm:

1. Start the program.
2. Read the student's area of interest.
3. Remove extra spaces and convert the input to lowercase.
4. Check whether the interest is ai.
5. If the interest is ai, recommend Artificial Intelligence.
6. Otherwise, check for data or web and recommend the corresponding course; if no match is found, display a message.
7. Display the recommended course and stop the program.

Pseudocode:

```
BEGIN
  INPUT interest
  CONVERT interest to lowercase
  IF interest = "ai"
    course = "Artificial Intelligence"
  ELSE IF interest = "data"
    course = "Data Science"
  ELSE IF interest = "web"
    course = "Web Development"
  ELSE
    course = "No matching course found"
  END IF
```

DISPLAY recommended course

END

Program:

```
interest = input("Enter your interest (AI, Data, Web): ").strip().lower()
if interest == "ai":
    course = "Artificial Intelligence"
elif interest == "data":
    course = "Data Science"
elif interest == "web":
    course = "Web Development"
else:
    course = "No matching course found"
print("Recommended Course:", course)
```

Input:

```
PS C:\Users\ANVES\Desktop\sep-dec\KED\LAB> & C:/Users/ANVES/AppData/Local/Programs/Python/Python313/python.exe c:/Users/ANVES/Desk
top/sep-dec/KED/LAB/3.py
Enter your interest (AI, Data, Web): data
```

Output:

```
Recommended Course: Data Science
```

Result:

Thus, a knowledge-based course recommendation system was implemented successfully using Python decision rules.